

Pokhara University
Faculty of Management Studies

Course Code: CMP 275

Course Title: **Computer Architecture and Microprocessor**

Nature of the course: Theory + Practical

Year: Second

Level: Bachelor

Semester: IV

Full marks: 100

Pass marks: 45

Credit Hrs: 3

Total periods: 48 hours

Program: BCSIT

1. Course Description:

This course aims at providing fundamental knowledge about computer architecture, instruction cycle, components of microprocessor, Intel 8085 and assembly programming.

2. Course Objectives:

The main objective of this course is to provide basic knowledge of components of microprocessor, block diagram and assembly language programming using Intel 8085, SAP1 and SAP2 computer architecture, timing diagrams, instruction cycles, machine cycles, control unit, central processing unit, RISC, CISC, Direct Memory Access, interrupts, serial and parallel interfaces to students of Bachelor Degree.

3. Course Contents:

Specific Objective	Contents
- To provide the basic history of computer - To provide the basic knowledge of stored program. - To provide the basic information about processor Components. - To provide the basic knowledge of internal communication of computer System - To provide the basic knowledge of SAP -1 and SAP- 2 Architecture	Unit 1: Introduction to Microprocessor (9 Hours) 1.1 Introduction and History of Microprocessors, Basic Block Diagram of a Computer, Bus Organization with Microprocessor Based System. 1.2. Stored Program Concept and its Processing Cycle, Microprogrammed and Hardwired control unit. 1.3. Microprocessor Components: Registers, ALU, Control and Timing, System Buses (Address, Data, Control), Microprocessor System with Bus Organization 1.4. SAP-1 Architecture: Block Diagram, and Function of each Block SAP-1 Instructions: LDA, ADD, SUB, OUT, HLT Fetch and Execution Cycle of SAP-1 Instructions with Timing Diagram Fetch Cycle: Address State, Increment State,



	Memory State Execution Cycle of LDA only 1.5. SAP-2 Architecture: Block Diagram and Functions of each Block, Architectural Differences with SAP-1 Bidirectional Registers Flags
• To provide the basic knowledge of Block diagram of 8085 microprocessor • To provide the basic knowledge of PIN • To provide the basic knowledge of ALL command • To provide the basic knowledge of ALL program with arithmetic's operation.	Unit 2: Intel 8085 (8 Hours) 2.1. Functional Block Diagram, Pin Configuration, Description of each Block: Registers, Flag (Description of each Flag), Data and Address Bus including Bidirectional Address/Data Bus, Timing and Control Unit, Interrupts (Introduction Only), Instructions: Op-Code and Operands, Addressing Modes, Instructions and Data Flow 2.2. 8085 Instructions: Data Transfer Instructions, Arithmetic and Logic Instructions, Branching and Machine Instructions 2.3. Basic Assembly Language Programming using 8085 Instruction Sets Addition, Subtraction, Multiplication and Division, Simple Sequence Programs, Array Searching and Sorting using Branching and Looping
	Unit 3: Computer Architecture Basics (3 Hours) 3.1. History of Computer Architecture Overview of 3.2. Computer Organization Architecture of Basic Computer: Register organization, Common Bus System Instruction Format, Register Stack and Memory Stack
• To provide the basic knowledge of Arithmetic's and logical operations.	Unit 4: Micro Operations (3 Hours) 4.1. Arithmetic Micro Operations: Addition, Subtraction, Increment, Decrement, Hardware Implementation 4.2. Logic Micro Operations: AND, OR, NOT, NAND, NOR, XOR, Selective Set, Set(preset), selective Complement(toggling) Insert, Hardware Implementation 4.3. Shift Micro Operations: Logical, Circular and Arithmetic, Arithmetic Logic Shift Unit



	Control Unit and Central Processing Unit
- To provide the basic knowledge of Control Unit structure. - To provide the basic knowledge of Control unit function. - To provide the basic knowledge of computer architecture - To provide the basic knowledge of Types of microprocessors.	Unit 5: Control Unit and Central Processing Unit (9 Hours) 5.1. Control Unit of Basic Computer and Timing Signal (Hardwired Vs. Microprogrammed) 5.2. Micro-operation, Micro-instruction, Micro-program: Symbolic and Binary Micro-program (FETCH and ADD) 5.3. Data Transfer operations and Manipulation (Arithmetic, Logical, Shift) 5.4. Introduction to RISC and CISC (Basic Differences)
	Fixed point Computer Arithmetic
- To provide the basic knowledge of Data representation. - To provide the basic knowledge of complement's method. - To provide the basic knowledge of Hardware algorithm.	Unit 6: Fixed Point Computer Arithmetic (5 Hours) 6.1. Signed number Representation: Signed Magnitude, 1's Complement and 2's Complement Form 6.2. Addition and Subtraction (with Numerical Example), Addition and Subtraction with Signed Magnitude Data, Hardware implementation, Hardware Algorithm, Addition and Subtraction with Signed 2's Complement Data 6.3. Multiplication Algorithm: Hardware Implementation for Signed Magnitude Data, Booth Multiplication Algorithm (with Numerical Example) 6.4. Division Algorithm: Hardware Implementation for Signed Magnitude Data, Hardware Algorithm (Restoring Only)
	Input and Output Organization
- To provide the basic knowledge of I/O interface. - To provide the basic knowledge of I/O mapped. - To provide the basic knowledge of DMA - To provide the basic knowledge	Unit 7: Input and Output Organization (5 Hours) 7.1. Introduction to Peripheral Devices, I/O interface-I/O bus and Interface Modules, isolated versus Memory Mapped I/O 7.2. Direct Memory Access (DMA): Introduction, Basic DMA Procedures (DMA



of protocol.	controller only) 7.3. I/O Processor, Data Communication Processor: Character Oriented Protocol and Bit Oriented Protocol
	Memory Organization
- To provide the basic knowledge of Memory System. - To provide the basic knowledge of I/O function of Memory devices. - To provide the basic knowledge of Memory core architecture.	Unit 8: Memory Organization (5 Hours) 8.1. Hierarchy of Memory System 8.2. Primary Memory: RAM and ROM, Memory Address Map with examples of Address Decoding. Secondary Memory: Structure of Magnetic Disk 8.3. Virtual Memory: Concept, Address Mapping with Pages, Basic Idea about Page Fault and page Replacement 8.4. Memory Management Hardware: Segmented Page Mapping (Introduction) , Memory Protection
	Pipelining
- To provide the basic knowledge of pipelining. - To provide the basic knowledge of pipelining instruction.	Unit 9: Pipelining (4 Hours) 9.1. Concept of Pipelining and Flynn's Classification, Pipelining Example with Speed Up Ratio 9.2. Arithmetic Pipeline, Pipeline for Floating-point Addition and Subtraction 9.3. Instruction Pipeline: Four Segment Instruction Pipeline 9.4. Data Dependency, Handling of Branch Instruction

Laboratory Works:

The laboratory works should be carried out in 8085 trainer kit. The programming should include arithmetic operation, base conversion, conditional branching etc.

Kit

1. Data swap and data transfer programs
2. 8-bit addition and subtraction



3. 16-bit addition and subtraction
4. 8-bit division and multiplication

Kit/Simulator

5. Series generation: odd-even, multiples of integers
6. BCD to ASCII
7. Searching largest/smallest value in an array
8. Ascending/ Descending sorting of array
9. Program involving bitwise AND, bitwise OR, bitwise X-OR, RLC, RRC

Text Books:

1. Ramesh S. Gaonkar: Microprocessor Architecture, Programming, and Applications with 8085, prentice Hall
2. Morris Mano: Computer system Architecture, Third Edition, prentice Hall

Reference Books:

1. Malvino: Digital Computer system Electronics (An introduction to Microcomputers)
2. Douglas V. Hall: Microprocessor and Interfacing programming and Hardware, McGraw Hill

